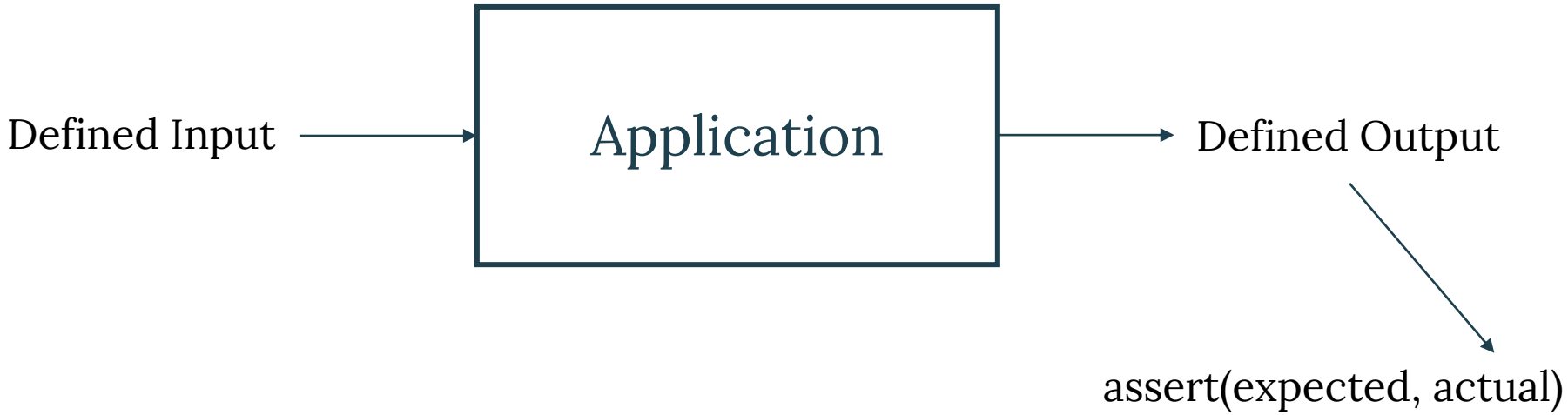


Model Evaluation

Tests


Defined Input
Application
Defined Output
assert(expected, actual)


LLMs are non-deterministic

Problem

- Prompt: „What‘s the capital of Germany?“
  - „Berlin“
  - „Berlin is the capital of Germany.“
  - „Berlin was the capital of Germany but now it‘s Bonn.“

Problem

- No single correct answer
- Examples
  - Summarizing a text
  - Classifying a support request
  - Generating an email

LLM as a Judge

- Ask LLM to evaluate a text
  - Requires special evaluation prompt
  - Use structured output
- Specialized models
  - Judge Arena: https://huggingface.co/spaces/AtlaAI/judge-arena

Typical Evaluations

- Correctness
- Relevancy
- Politeness
- Tone
- Company policies


Evaluation during Tests

Test Evaluation

- Based on *Evaluator* interface
- Standard *Evaluator*s
  - *FactCheckingEvaluator*
  - *RelevancyEvaluator*

Code Samples

Local Models

- Problem
  - Limited resources
  - Weak models

Local Models

- Options
  - Local inference server
  - Testcontainers
    - Docker Model Runner
    - Ollama in a Docker container

Code Samples


Costs & Complexity

Assertions

- Structured output
- Separate deterministic from non-deterministic
  - Business logic
  - Some Advisors
  - Some Tools
  - RAG retrieval

Test Model

- Use test models
  - Technical end-to-end tests
  - Advisor chains
  - RAG

Final Tests

- Always test like production
  - Same model
  - Same vector store
  - Same tools / MCP servers


Evaluation at Runtime

Runtime Evaluation

- Constraints
  - Time
  - Money / resources

Runtime Evaluation

- Based on
  - Recursive *Evaluator*
  - Support for multiple *ChatModel*s

Code Samples

Runtime Evaluation

- https://huggingface.co/learn/cookbook/en/llm_judge#3-improve-the-llm-judge
- https://spring.io/blog/2025/11/10/spring-ai-llm-as-judge-blog-post



Spring Community: Agent Judge

Agent Judge

- Agent-agnostic evaluation framework
- Judges based on
  - Command execution
  - LLM
  - File comparison
- Supports judge composition

Agent Judge

- https://github.com/spring-ai-community/agent-judge


Questions